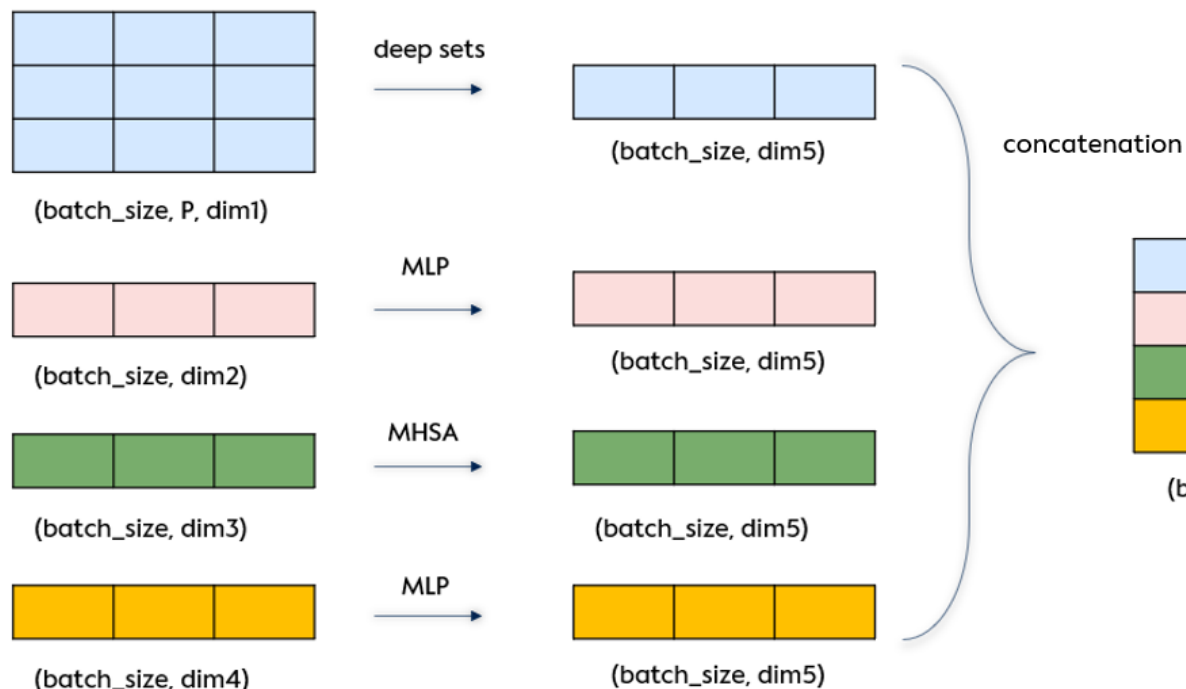


[3] Permutect: An Elegant Method for Somatic and Germline Variant Calling

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 *The author list is not final.



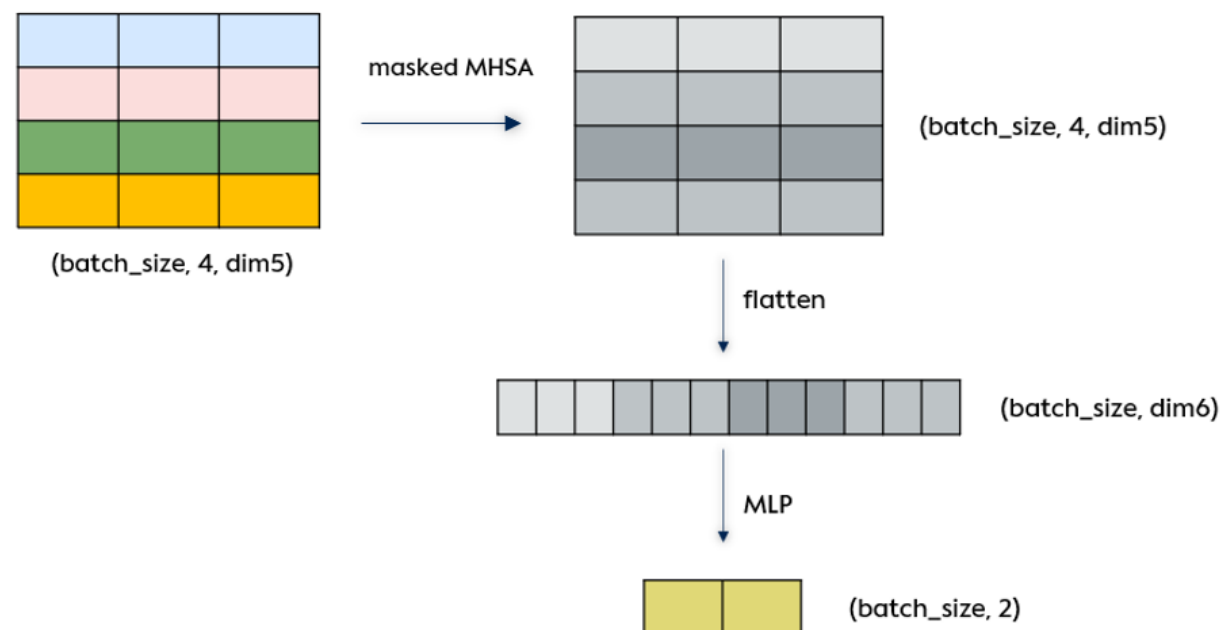
	learned features	handcrafted features
sample specific features	per read features as in [3]*	same features as the sample-specific features for the regression model
genomic context specific features	raw genomic context	same features as the context-specific features for the regression model

For visibility batch_size is set to 1 in this diagram.

Masked MHSA is done on 2nd dimension (4) to dissallow the attention of 1 tensor to another (e.g. The green one to itself).

The final MLP wrongly induces a meaning to the order of the elements in the flattened vector. But this is done intentionally to circumvent the potential oversmoothing effect of a permutation invariant aggregation function.

The two output nodes are the two parameters of a zero-inflated binomial model. The loss function is a zero-inflated binomial loss.



$$\epsilon_1, \epsilon_2 = M(\mathbf{X}, \theta) \quad L_{(\epsilon_1, \epsilon_2)} = \begin{cases} 0, & \epsilon_1 + (1 - \epsilon_1)(1 - \epsilon_2)^{n_i} \\ k, & (1 - \epsilon_1) \binom{n_i}{k} \epsilon_2 (1 - \epsilon_2)^{n_i - k} \end{cases}$$